

CoRoT-2b

R-0247

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_230710 · created 2026-07-17T06:02:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460135.8399 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.004 +/- 0.062
Transit depth (Rp/Rs)^2	0.0013 +/- 0.045 [%]
Orbital Inclination (inc)	89.81 +/- 2.03
Airmass coefficient 1 (a1)	4.53 +/- 3.01
Airmass coefficient 2 (a2)	-1.16 +/- 1.75
Scatter in the residuals of the lightcurve fit is	76.51 %
Best Comparison Star	None
Optimal Aperture	2.65
Optimal Annulus	3.5
Transit Duration (day)	0.0842 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.004 ± 0.062 vs 0.1667 (deviation 1.84σ)
Duration fit vs published	0.0842 d vs 0.095 d (deviation 1.76σ)
Mid-time O–C	2.3 min
Depth SNR	0.0
Residual scatter	76.51 %

Light curve

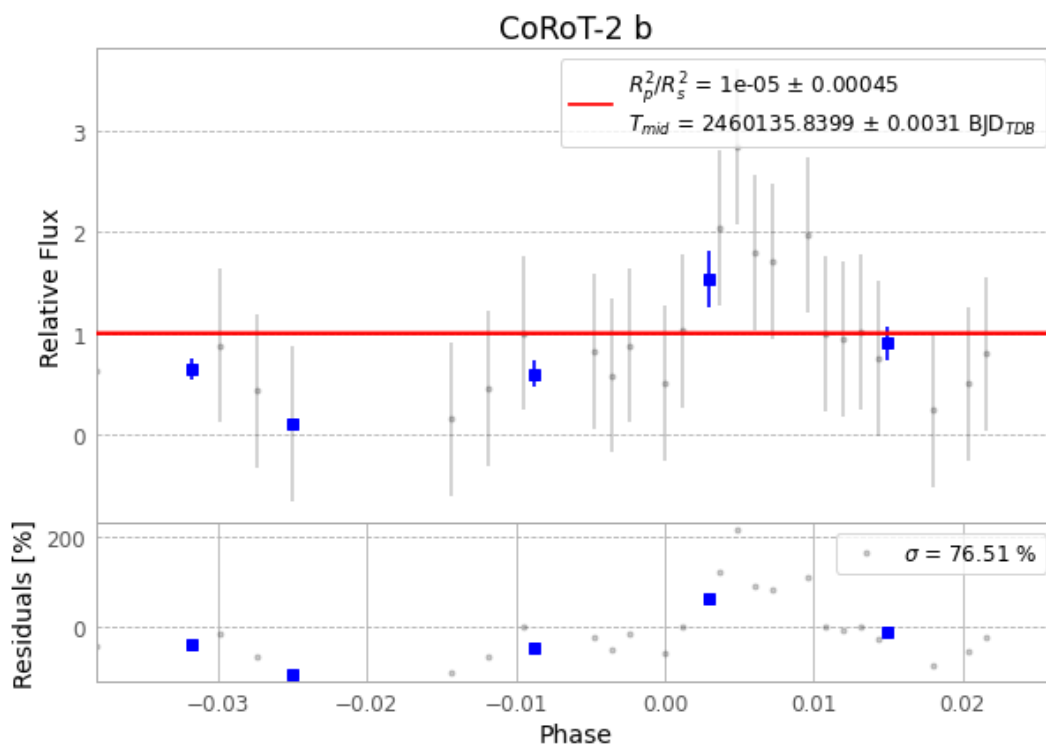


Figure 1 — FinalLightCurve_CoRoT-2 b_2023-07-09.png (SHA-256 c10a3a3b1651d4d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [537, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5f49f70686f273fe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

48 FITS frames (32 MB), CoRoT-2230710062115.FITS → CoRoT-2230710091215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-230710.log (e52ca4c274eb...), FinalLightCurve_CoRoT-2 b_2023-07-09.png (c10a3a3b1651...), FinalLightCurve_CoRoT-2 b_2023-07-09.csv (cdc6612bd7d0...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 06:02Z.